SOFC PLANT TWIN · REPLAY TEST AGAINST THE POC DATA · RECHECKED 4 NOV ON TWIN V16

Twin vs plant: how close is it?

Today's twin follows the real plant closely in every stage but one, and most of its buttons do what the plant's operators did. After a trip it now goes back to hot standby, as the plant usually did. Two rules still differ: the Load button opens the gas too early, and part load always ends in hot standby, which is stricter than the plant.

We fed the twin the plant's own knob settings minute by minute (air, gases, heaters, current) and compared what it computes. Then we pressed the twin's buttons and compared what they do with what the plant did over 39 days.

What changed in the twin today

Twin version	Change	Effect on this test
v14	Trip purge only from hot standby or part load; after the purge the twin cooled to shutdown.	Checked this morning (pages 7–8, first version).
v15	After a trip purge: back to hot standby. Part load always ends in hot standby (Shut down, gas off or current 0 in part load first unload to standby).	“After a trip” now **matches** the plant. New row: “Leaving part load”.
v16	The air-off stage is now called **Shutdown / Emergency stop**; button “Shut down / E-stop”.	Name only. Behaviour is the same.
check	Air heater (EAPH) box readings compared with the plant.	New row: the box shows look-up values, not the air-in the twin computes.

The heat balance itself did not change. Rerunning the 8 replays on v16 gives the same stack temperature, air-in and stage name in every minute (difference 0.0 °C).

WHAT THE TWIN COMPUTES · plant knobs fed in

Stack temperature, 8 replays 17.5 °C avg; 6 of 7 stages within 5 °C	MATCHES
Stage name right in 99 % of minutes	MATCHES
Part load, 10 plant runs 4.2 °C avg	MATCHES
Air going into the stack within 4 °C; +19 °C in air-flow cooling	CLOSE
110 sensor readings typical miss 6 % of range	CLOSE
Air heater box readings look-up; power flat inside a stage	DIFFERS
Fast cooling, 9000 ln/min air 75 °C too warm on average	DIFFERS

WHAT THE TWIN'S BUTTONS DO · vs plant practice

Cold start button 0.2–0.6 h ahead of the plant	MATCHES
Fuel switch → part load 10 min; plant median 9 min	MATCHES
Trip purge only from standby / load fits 5 of 6 plant purges	MATCHES
After a trip: back to hot standby new today; plant did this 4 of 6 times	MATCHES
Load button: when gas opens 506 °C; plant waits for ≈600 °C	DIFFERS
Leaving part load twin always to standby; plant 2 of 10	DIFFERS
Holding part load for hours keeper; plant never ran over 44 min	NO PLANT DATA

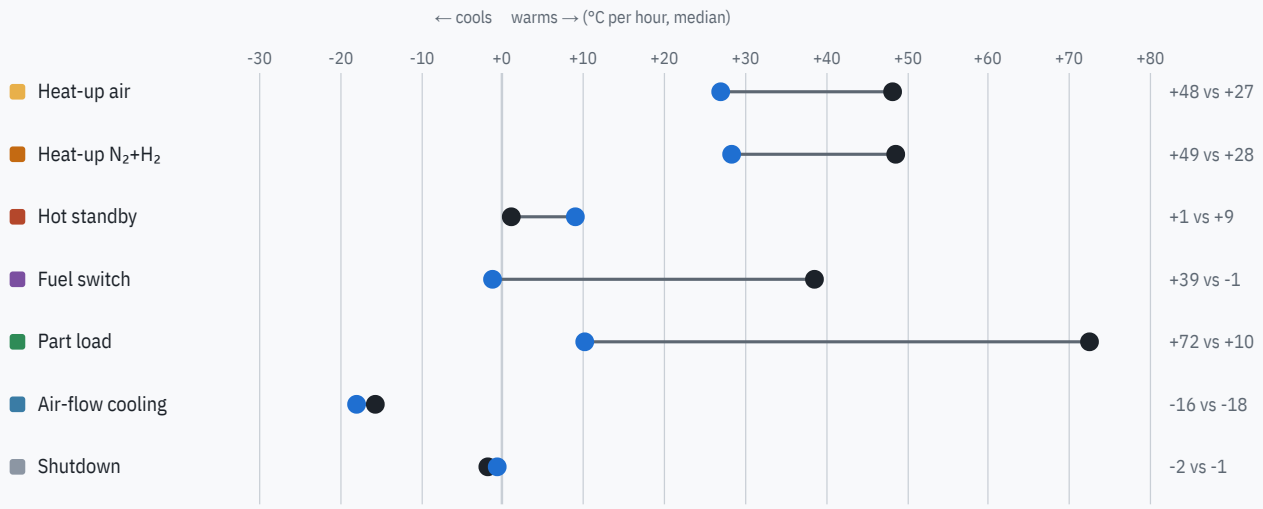
Green = matches the plant, blue = close, red = differs, grey = the plant data cannot tell. Plant data: 39 days, 8 replay stretches (6,844 minutes), 10 part-load runs, 19 fuel switches, 9 trips.

Page 7 · temperatures, stages and readings →

Page 8 · buttons, rules and decisions →

Where we started: the first test, 3 Nov

Everything below is the first replay test, before the fixes on pages 4–6. Back then the twin got the stage order right but heated the stack too slowly: whenever the plant warmed up (start-up, fuel switch, part load) the twin fell 40–140 °C behind. In slow cooling it was within about 10 °C.



Black dot = how fast the real stack warmed or cooled in that stage. Blue dot = the twin under the same settings. The longer the grey line, the bigger the miss. Median over 8 replays, 6,844 plant minutes. First test, 3 Nov.

Stage by stage, first test

Stage	Minutes compared	Plant °C/h	Twin °C/h	Twin miss, avg °C	Stage name righ
Heat-up air	575	+48	+27	-17	95
Heat-up N_2+H_2	929	+49	+28	-136	100
Hot standby	1302	+1	+9	-62	26
Fuel switch	53	+39	-1	-10	98
Part load	99	+72	+10	-44	100
Air-flow cooling	2355	-16	-18	+40	100
Shutdown	530	-2	-1	+7	100

“Twin miss” is twin minus plant stack temperature, averaged over the minutes in that stage. Hot standby scores low on “stage name right” because the twin is still heating when the plant has already settled, so the twin calls it heat-up.

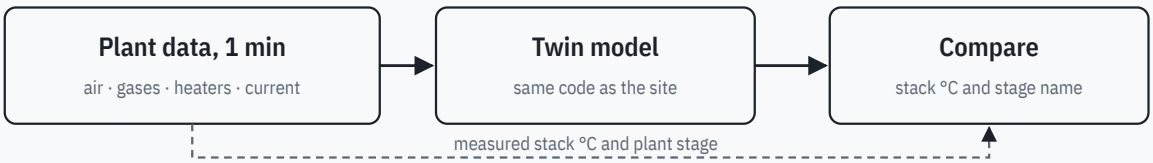
Page 2 shows all eight replays as curves. Page 3 explains why the twin warms too slowly.
Page 4 lists what to change.

IF YOU REMEMBER ONE THING

Today's twin computes the plant's temperatures well (17.5 °C average, 6 of 7 stages within 5 °C), and after a trip it now returns to hot standby like the plant. What still differs: when the gas opens, the fixed way out of part load, and the air-heater box readings.

Eight replays, plant against twin

Each chart is one real stretch of plant data. The twin started at the plant's stack temperature and then received the plant's knob settings every minute.

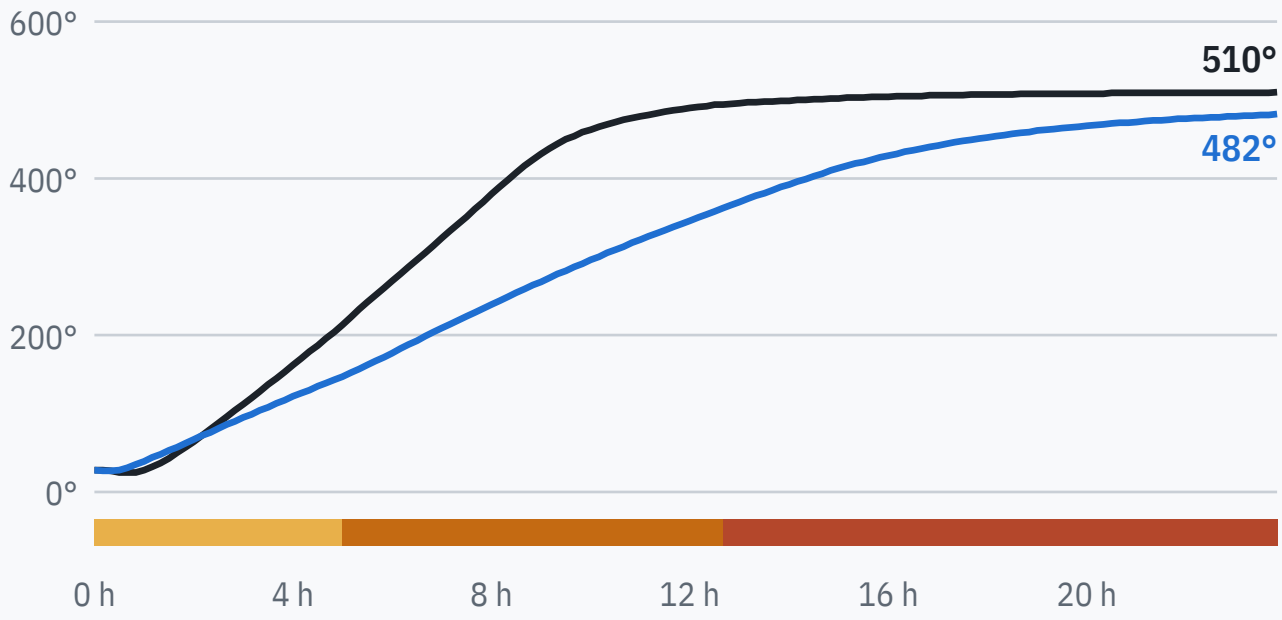


The twin is not told the plant's temperature after the first minute. Any gap that opens up is the model's own error.

Plant (measured) Twin, fed the plant's knob settings

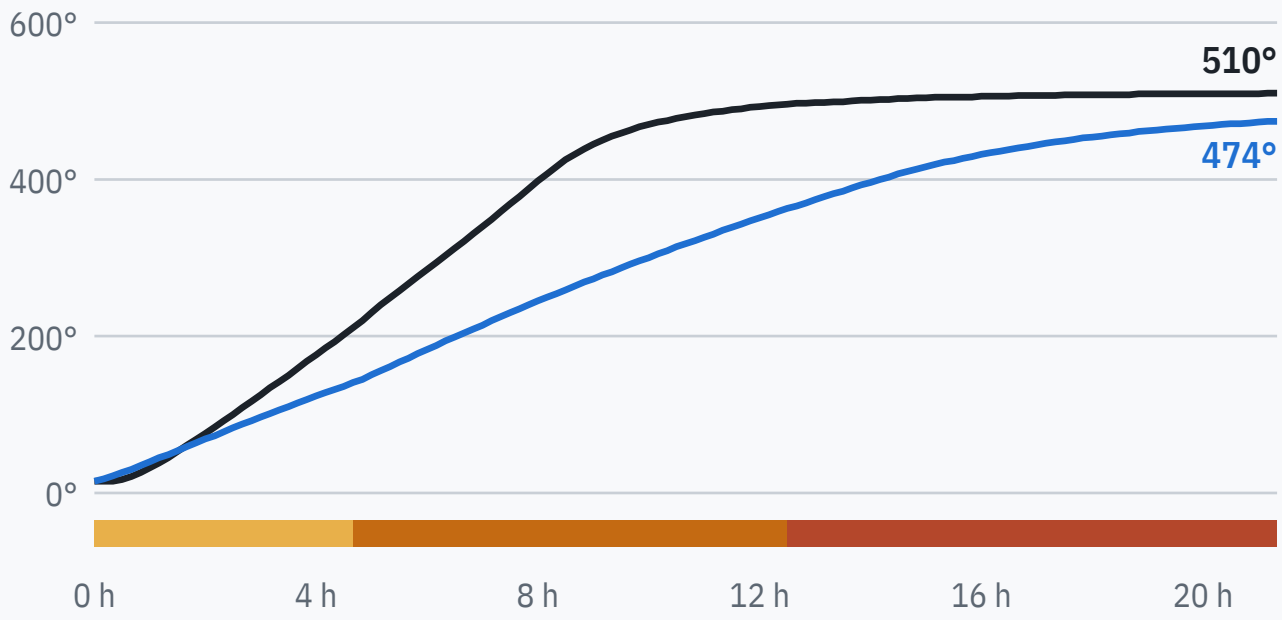
Heat-up air Heat-up N_2+H_2 Hot standby Fuel switch Part load Air-flow cooling Shutdown Trip: purge

Cold start · 1 Mar



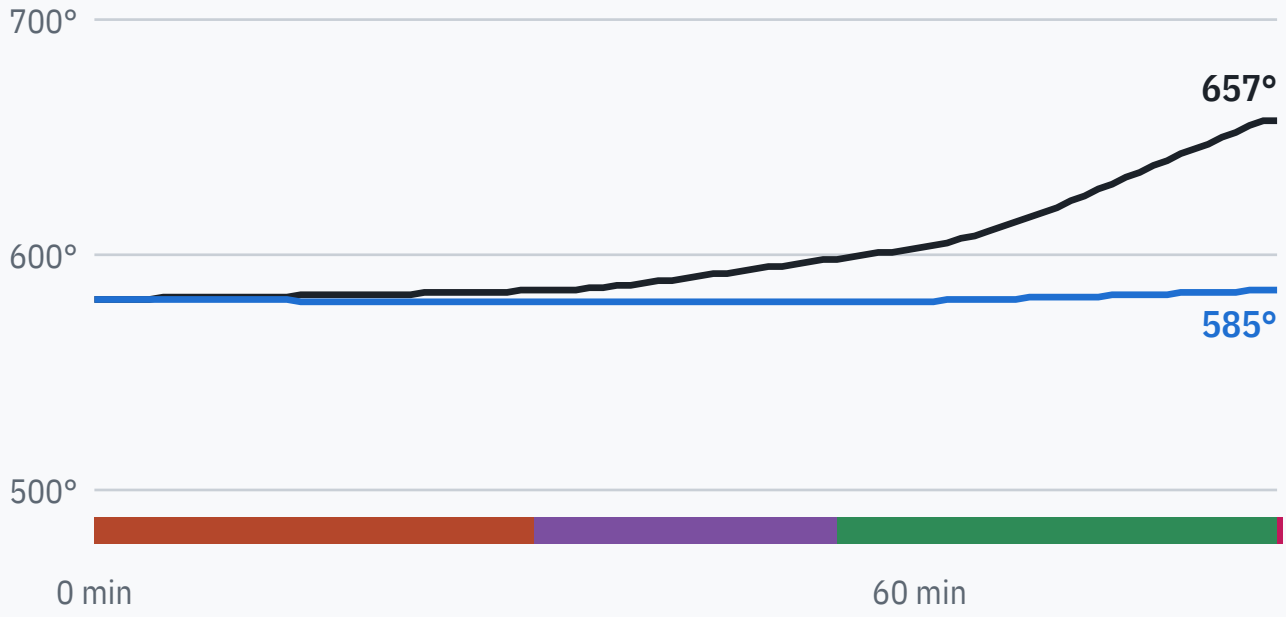
Plant reaches 490 °C after 12.0 h. The twin never gets there within the day (ends at 482 °C).

Cold start · 13 Feb



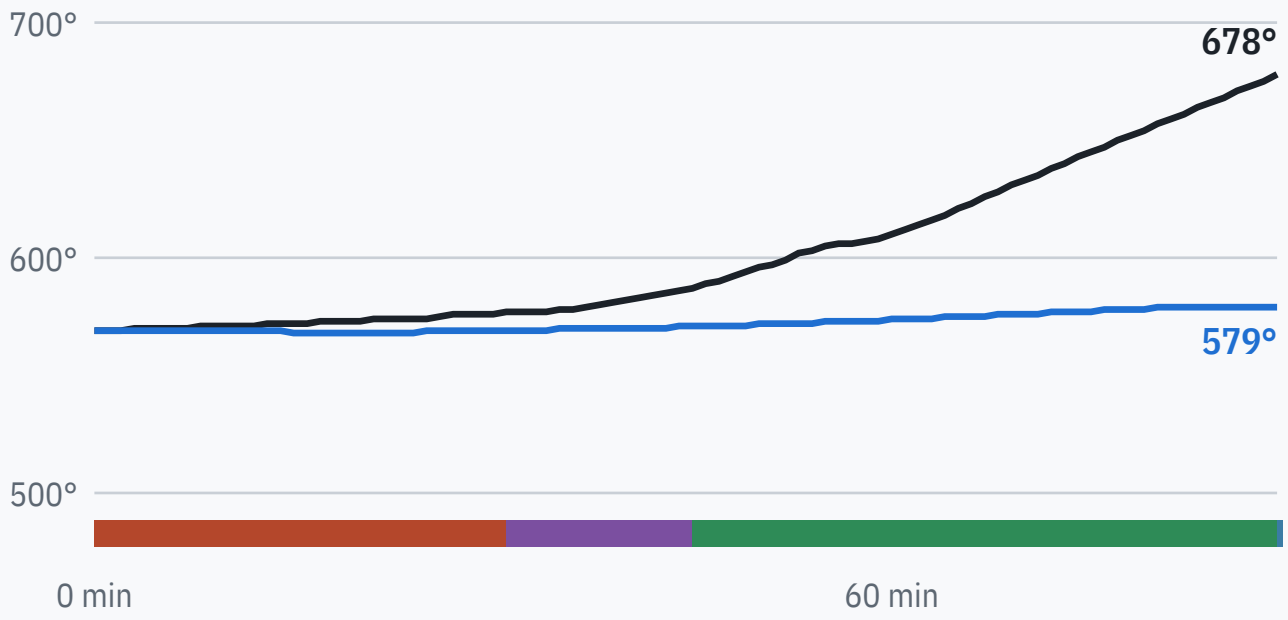
Same picture on the second start: plant 11.7 h to 490 °C, twin never.

Fuel switch → part load · 24 Mar



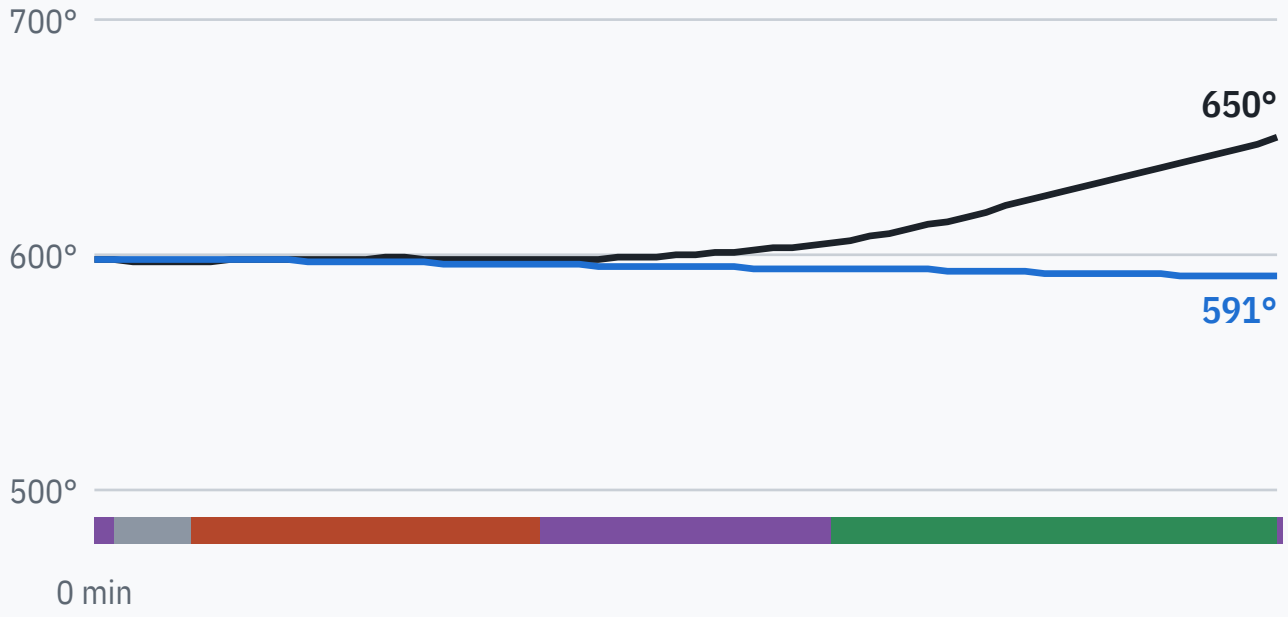
Plant climbs 581 → 657 °C through fuel switch and part load. The twin stays near 585 °C.

Fuel switch → part load · 25 Mar



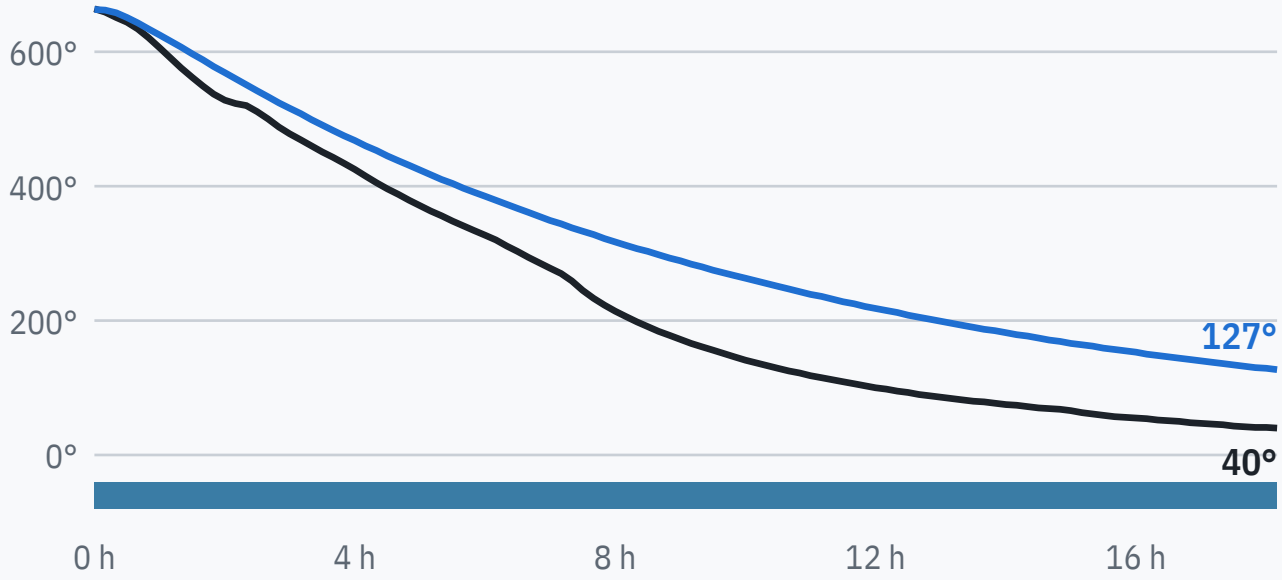
Plant +109 °C over these 90 min; twin +10 °C.

Fuel switch → part load · 30 Mar



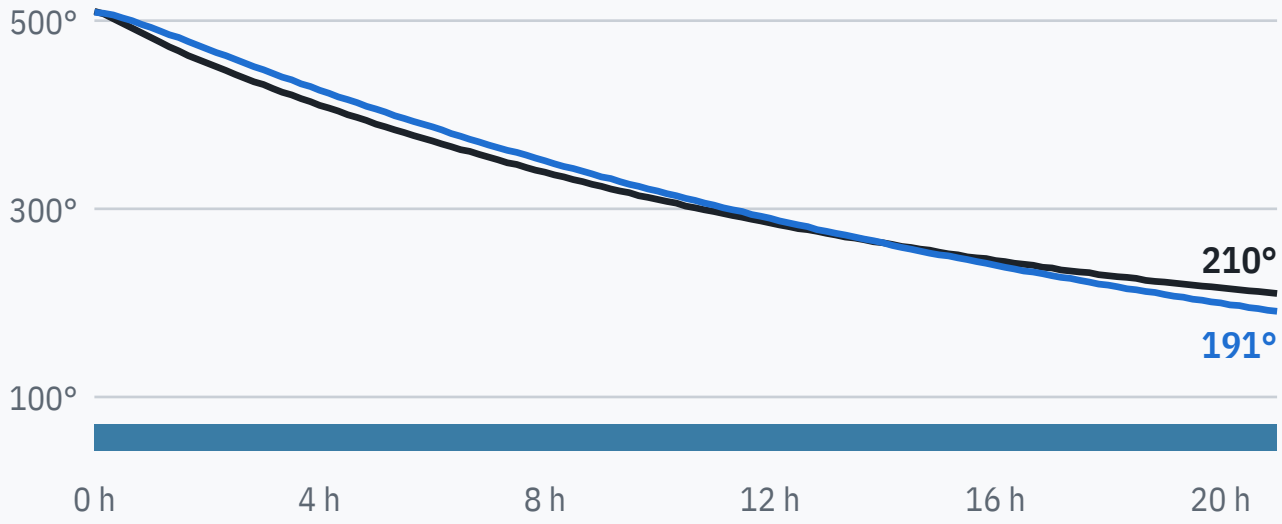
Current under 3 A and the plant still warms 110 °C/h, so the heat comes from the gas, not the current.

Air-flow cooling, 9000 air · 25 Mar



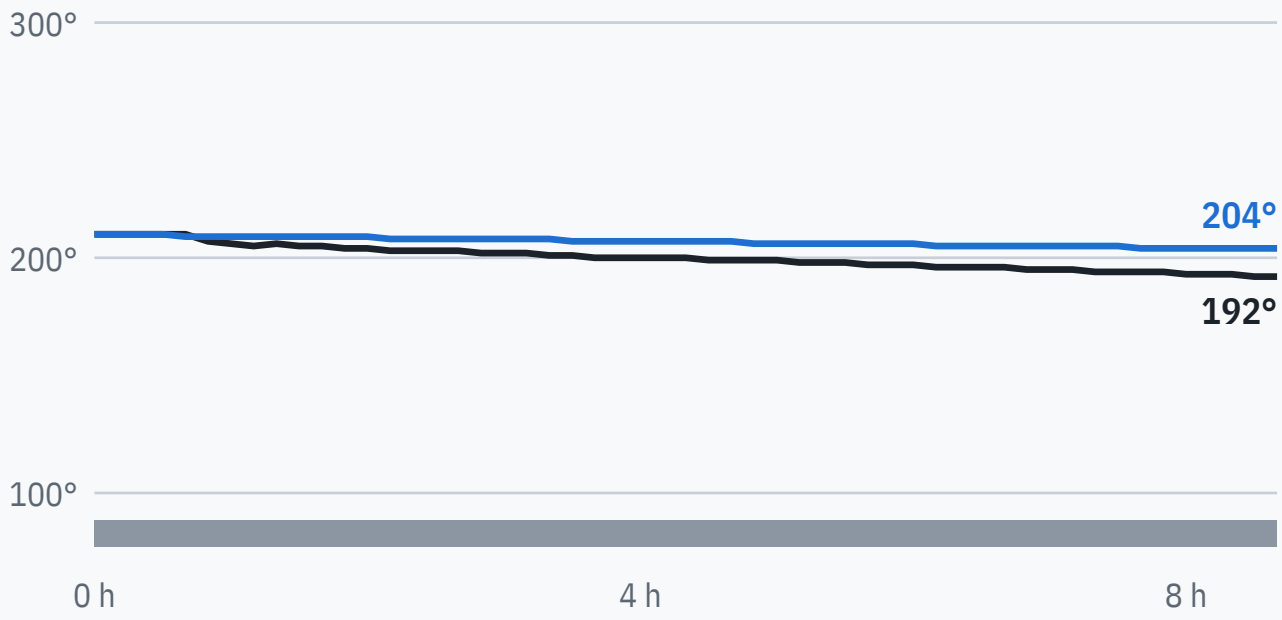
Cooling with 9000 ln/min of air: the plant cools to 40 °C, the twin only to 127 °C.

Air-flow cooling, 6000 air · 16 Feb



Cooling with 6000 ln/min of air: twin within 11 °C on average.

Shutdown, air off · 17 Feb



Air off, everything off: twin within 8 °C.

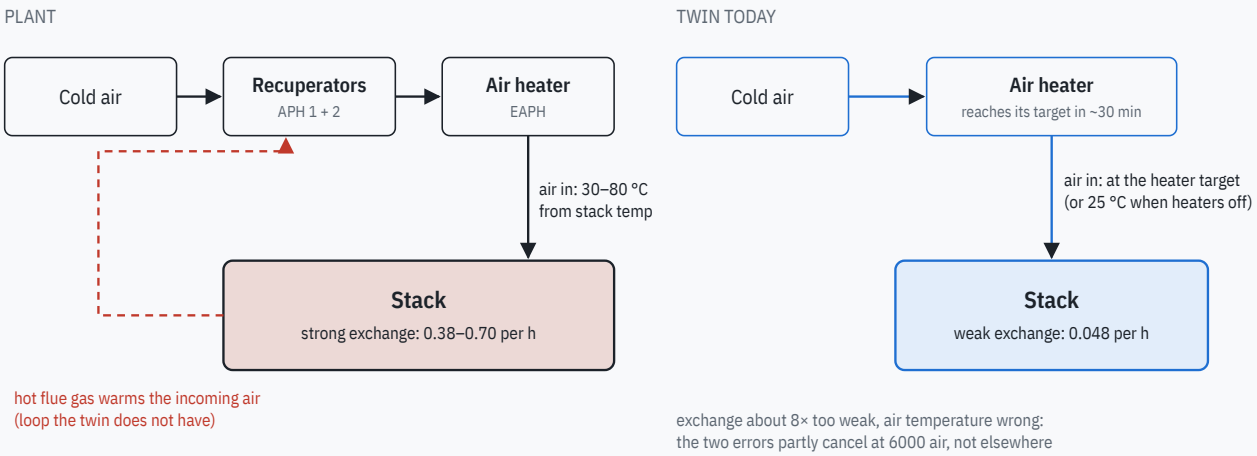
IF YOU REMEMBER ONE THING

Cooling and shutdown curves sit on top of each other. Start-up and fuel switch / part load curves split apart because the twin warms too slowly.

Why the twin warms too slowly

Two pieces of physics are missing: **how the plant re-uses its own exhaust heat**, and **the heat that natural gas gives off**.

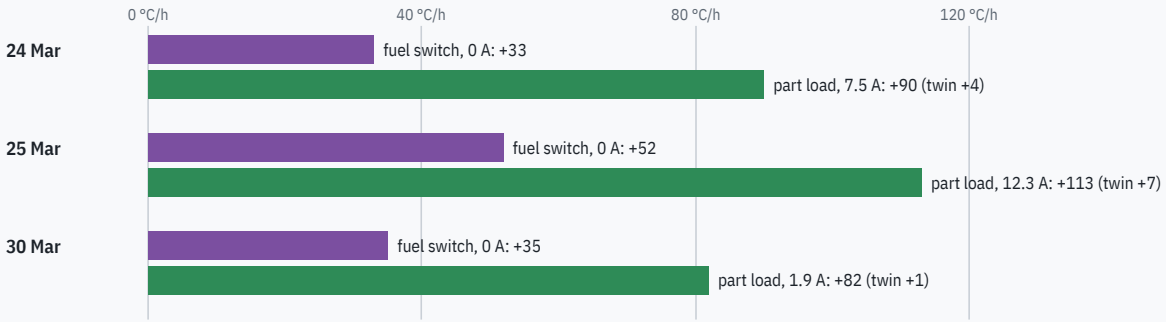
1. The air path



Measured on the cooling replays (heaters off): the stack trades heat with its air about 8 times faster than the twin assumes (0.38 per hour at 6000 ln/min, 0.70 at 9000; twin 0.048 and 0.10). The plant's air arrives only 30–80 °C away from the stack because the recuperators pre-warm it with the flue gas.

When we fed the twin the plant's *measured* air-in temperature, it got **worse**, not better (cold start ends 470 vs 510 °C; 9000-air cooling ends 207 vs 40 °C). So the right air temperature alone does not help. The heat exchange itself is too weak, and today the twin only lands close because a wrong air temperature cancels it out.

2. Heat from the gas



During fuel switch the current is zero, yet the stack warms 33–52 °C per hour. That heat comes from the natural gas and POx air (partial oxidation in the reformer and burning in the afterburner). The twin has no such term: gas only changes the stage name. In part load the twin adds 0.6 °C/h per amp, which is 1–7 °C/h against 82–113 measured.

3. A smaller point: the start-up target

The twin's “Cold start” button sets the air heater to 650 °C. The plant used **500 °C** on both start-ups. With the plant's 500 °C the twin never reaches 490 °C in a day. The 650 °C in the button was hiding the missing heat.

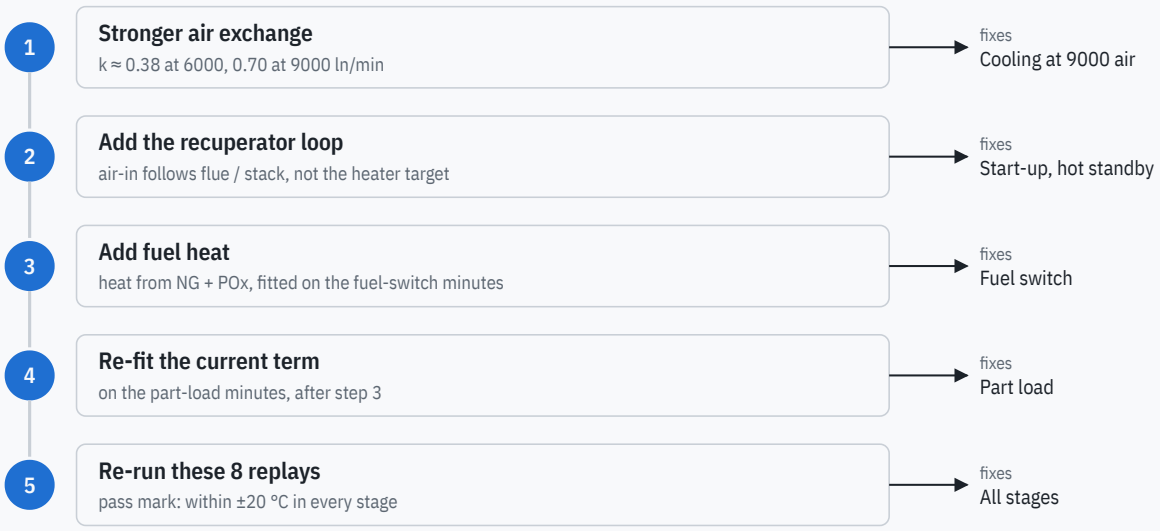
Cold start milestone	Plant (2 starts)	Twin button (650 °C)	Twin with plant settings (500 °C)
Forming gas at 210 °C	4.7–5.0 h	4.6 h	6.9–7.0 h
300 °C	6.3–6.6 h	7.1 h	10.0–10.2 h
490 °C	11.7–12.0 h	13.7 h	not reached

IF YOU REMEMBER ONE THING

The plant warms itself with its own exhaust and with its fuel. The twin has neither, so it only matches the plant when nothing is warming up.

What to change in the twin

Five changes, in this order. **All five are now done** (page 5). Each one is fitted on the same 39 days and checked on these 8 replays.



Order matters: fix the air first (it moves every stage), then add the fuel heat, and only then re-fit the current, so the current term does not absorb heat that really comes from the gas.

What already works and stays

Part	Result
Stage names in cooling, shutdown, fuel switch, part load	98–100 % right
Slow cooling with 6000 ln/min air	within 11 °C on average
Cooling with air off	within 8 °C on average
Fuel switch → part load after 10 min (added today)	matches the all-days median of 9–10 min; in these 3 replays the plant took 14–22 min

Decisions for you

A. Pass mark for the replay test: ±20 °C average miss per stage (suggested), or stricter?

B. Change the “Cold start” button to the plant's 500 °C target now (honest, but slower until fixes 1–2 are in), or keep 650 °C until then?

C. Fuel switch → part load wait: keep the fixed 10 min, or draw it from the plant spread (5–30 min)?

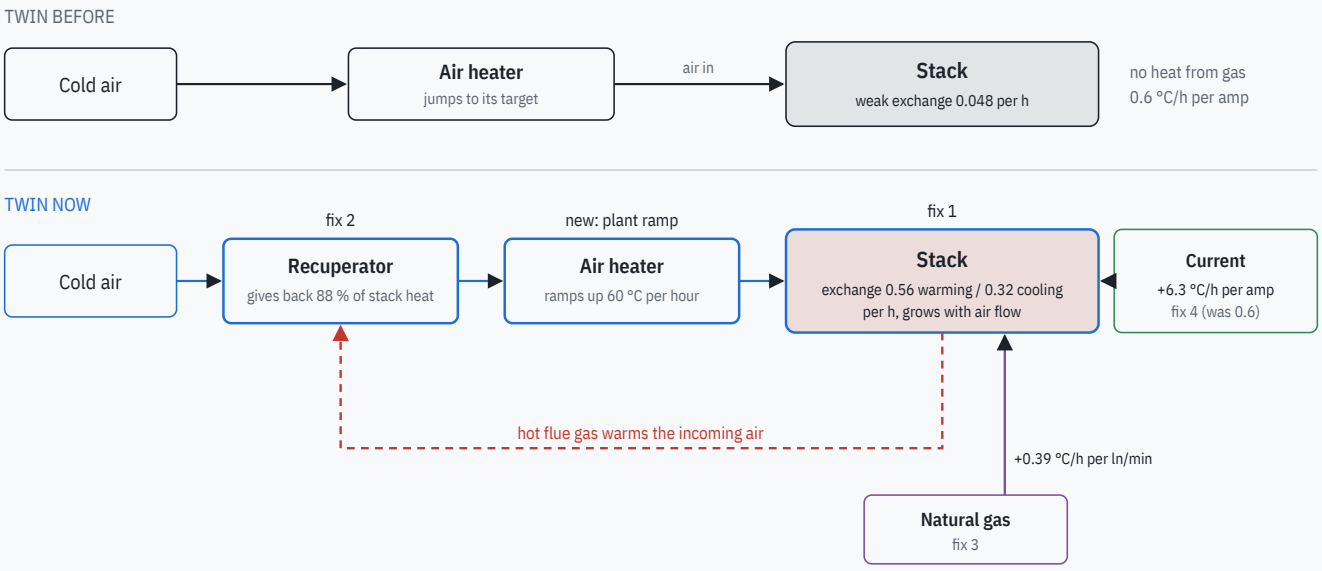
```
Replay scripts: session scratchpad cmp/ — extract.py (plant episodes), replay.js (runs twin_2d/app.js model), score.py, summ.py
```

> IF YOU REMEMBER ONE THING
>
> Fix the air exchange first, then add the fuel heat. After that the replay test tells us, stage by stage, whether the twin is good enough.

What we changed in the twin

All five fixes from page 4 are in. The twin now has the recuperator loop, an air heater that ramps like the plant's, heat from the gas and a stronger current term.

Every number was fitted on the 39 days of plant data **with the 8 replay stretches left out**, so page 6 is a fair test.



Top: the twin before. Bottom: the twin now. The red dashed loop is the recuperator: hot exhaust pre-warms the incoming air, so the air reaches the stack close to stack temperature. The coloured arrows are the two new heat sources: natural gas and electric current.

Plan vs what the data said

#	Fix	Plan (page 4)	Fitted from the plant	Note
1	Stronger air exchange	0.38 at 6000, 0.70 at 9000	0.56 per h when air warms the stack, 0.32 when it cools it; grows with air flow (0.58 cooling at 9000)	Warming and cooling need different values: the insulated hot box gives heat back while cooling.
2	Recuperator loop	air-in follows flue / stack	air-in = 88 % of the way from room to stack temperature (83 % at 8000+ air)	Checked against the measured air-in: 0–4 °C median miss per stage.
2b	Air-heater ramp (new)	not in the plan	the plant raises the air-in 60 °C per hour (9 of 12 heat-ups: 59–64)	Without the ramp the twin warmed too early at the start.
3	Fuel heat	fit on fuel-switch minutes	+0.39 °C/h per ln/min natural gas (≈ +31 °C/h at 80)	Plant fuel switch warms +25 °C/h median on the fitting days.
4	Re-fit current	after step 3	+6.3 °C/h per amp per string (was 0.6)	Part load: plant +84, fit +89 °C/h median.
5	Re-run 8 replays	pass mark ±20 °C	done, see page 6	

Also changed

Plant-speed shortcut removed. Until now, in the four warming stages the twin copied the plant's average warming speed from a table. With the physics fixed, the heat balance is better in all four stages (page 6), so the twin uses one heat balance everywhere again.

“Cold start” button uses the plant's settings (decision B on page 4): air heater 300 °C, raised to 500 °C when forming gas starts at 210 °C. It was 650 °C, which only hid the missing heat.

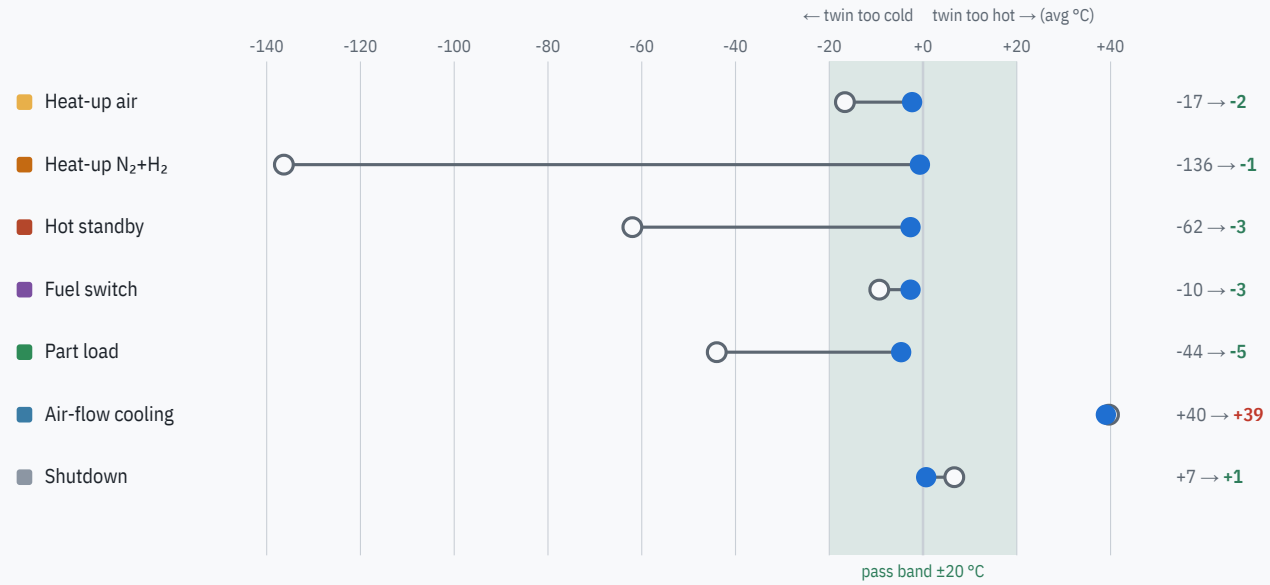
`Code: twin_2d/app.js (model block at the top), twin_2d/README.md · fits: session scratchpad fix/ (fit5.py, airramp.py) · replay: replay.js, score.py`

> IF YOU REMEMBER ONE THING

> The twin now warms itself the way the plant does: with its own exhaust heat, with its fuel and with its current.

After the fixes: within ±20 °C almost everywhere

Average miss dropped from 56 °C to 17.5 °C. Six of the seven stages are now inside the ±20 °C pass mark. Only cooling at 9000 ln/min air still misses.



Average gap between twin and plant stack temperature per stage (twin minus plant), over the same 6,844 plant minutes as page 1. Green band = pass mark ±20 °C.

Each replay

Replay	Before, avg °C off	Now
Cold start 1 Mar	76.4	**2.6**
Cold start 13 Feb	85.2	**3.1**
Standby → power 24 Mar	19.1	**2.0**
Standby → power 25 Mar	29.5	**2.8**
Standby → power 30 Mar	14.7	**6.8**
Cooling, 9000 air	81.0	**74.5**
Cooling, 6000 air	10.2	**9.6**
Shutdown, air off	6.9	**0.8**

The plant-speed shortcut (twin v8) scored 23.9 °C overall; the fitted physics now scores 17.5.

Cold start milestones

Stack reaches	Plant	Twin before	Tw
210 °C (forming gas)	4.7–5.0 h	6.9–7.0 h	**4.8**
300 °C	6.3–6.6 h	10.0–10.2 h	**6.4**
490 °C	11.6–12.0 h	never	**11.6–**

Twin given the plant's knob settings minute by minute. The “Cold start” button alone: 4.5 h, 6.1 h, 11.4 h, then settles at 506 °C (plant 509).

The one miss left: cooling with 9000 ln/min air

The twin cools too slowly: after 18 hours it shows 145 °C where the plant had 41 °C. When we feed it the plant's *measured* air-in temperature, it follows the plant to within about 25 °C. So the stack's heat balance is right. What is off is the recuperator at high air flow.

On this stretch the incoming air was only 72–82 % of the way to stack temperature. On the other days with high air flow it was 79–89 % (the twin uses 83 %). We did not tune the twin on this test stretch.

Rechecked on 4 Nov with twin v14 and again with v16: the same replays still give 17.5 °C. See page 7. The open decisions are now collected on page 8.

Decisions for you (3 Nov)

A. Pass mark: we used ±20 °C average per stage. Keep it, or make it stricter now that six stages are within 5 °C?

B. Done: the “Cold start” button now uses the plant's 300 → 500 °C. Keep it?

C. Fuel switch → part load wait: still a fixed 10 min (unchanged).

D (new). 9000-air cooling: let the recuperator fall to about 78 % at 9000 air (would pass, but is tuned on one stretch), or wait for more forced-cooling days?

IF YOU REMEMBER ONE THING

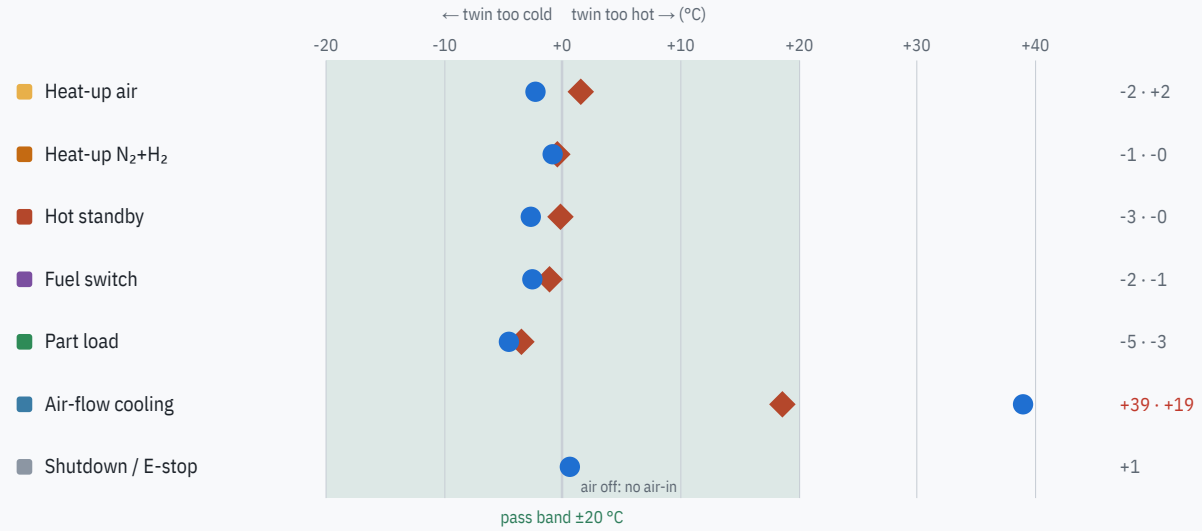
With its own exhaust heat, its fuel and its current, the twin now follows the plant in every stage except one: fast forced cooling at 9000 ln/min air.

Today's twin: temperatures, stages and readings

Given the plant's own settings, today's twin is 17.5 °C from the real stack on average, the same as after the 3 Nov fixes. The 4 Nov changes (stack keeper, trip rules, air-in held at the heater target, part load → hot standby, the new stage name) left the replays as good as before. Fast cooling at 9000 ln/min air is still the one miss.

Rerun on twin v16: every minute of the 8 replays gives the same stack temperature, air-in and stage name as v14. The v15 and v16 changes only touch what the twin does on its own after a trip or when part load ends, and the name of the air-off stage (now “Shutdown / Emergency stop”). In a replay the plant's own settings drive every step, so these rules never fire.

Stack and air-in temperature, stage by stage



● stack temperature, average twin – plant ◆ air going into the stack (TIRC 2.06), typical twin – plant

Same 8 plant stretches and 6,844 minutes as pages 1 and 6, run through the twin code as it is today. In air-flow cooling the twin's incoming air is about 19 °C too warm, so its stack cools too slowly. That is the 9000-air stretch from page 6.

Stage	Minutes	Stack, avg °C off	Air-in, typical °C off	Stage name right
Heat-up air	575	-2.3	+1.6	95 %
Heat-up N_2+H_2	929	-0.8	-0.4	100 %
Hot standby	1,302	-2.7	-0.1	99 %
Fuel switch	53	-2.5	-1.0	98 %
Part load	99	-4.5	-3.4	100 %
Air-flow cooling	2,355	+38.9	+18.6	100 %
Shutdown / Emergency stop	530	+0.6	–	100 %

Changes since 3 Nov: fuel switch −2.8 → −2.5 °C, part load −4.8 → −4.5 °C, the 30 Mar replay 6.8 → 6.1 °C. Everything else is unchanged to 0.1 °C. The plant's air-heater target is smoothed over 5 minutes, as on 4 Nov, because it blips for one minute at restarts.

The 10 plant part-load runs

All ten real part-load runs (8–44 min each), replayed with the plant's settings: **4.2 °C average miss**, both before and after the 4 Nov changes.

Largest miss: 30 Mar, 26 °C too cold at the end. That run had very little current (2.8 A per string), yet the plant still warmed fast.

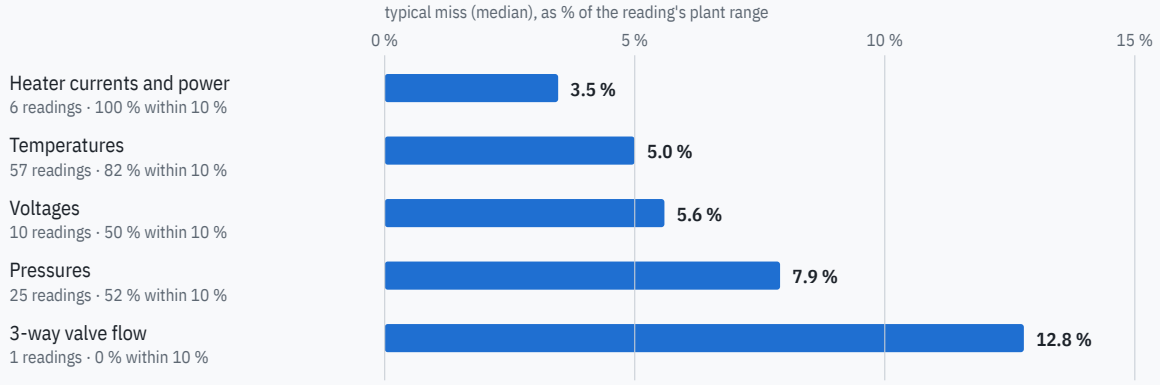
Stage names

The twin names the right stage in **99 % of the minutes**. On page 1 hot standby was only 26 % right, because the old twin was still heating when the plant had settled.

The only misses: 31 minutes at the very start of a heat-up, called air-flow cooling because the heaters were not yet on.

The 110 sensor readings on the twin screen

The twin shows 110 readings besides the stack temperature. Most come from a look-up: for each stage, a straight line against the stack temperature, learned from the 39 days. We compared them minute by minute with the real sensors in the same 8 stretches. 99 readings change in these stretches; the other 11 stay flat there and are left out.



Overall: typical miss **5.8 %** of a reading's range; 72 % of readings are within 10 %, 86 % within 20 %. Example: a heater temperature that spans 550 °C misses by about 25 °C.

By stage

Plant stage	Typical miss
Hot standby	0.8 %
Heat-up air	2.2 %
Heat-up N_2+H_2	2.4 %
Shutdown / Emergency stop	2.8 %
Air-flow cooling	6.1 %
Part load	8.5 %
Fuel switch	9.2 %

Steady stages are almost exact. Fuel switch and part load are short, fast-moving stages with few minutes to learn from.

Weakest readings

Tag	What		Miss of range
PIR 1.4.1	After booster	4.67 bar	34 %
PIR 1.2.1	POx air supply	0.19 bar	31 %
PIR 7.1.2	APH inlet exhaust 2	2.91 mbar	28 %
TIR 3.02	Ejector main stream	9 °C	27 %
TIR 1.3	Purge gas	5 °C	26 %
PIR 1.1.3	Air stream 2 supply	0.22 bar	26 %

These are mostly supply-side readings (gas and air supply pressures, inlet gas temperatures). They follow the supply and the weather, not the plant stage, so a stage look-up cannot follow them. The misses are small in plain units.

Fair warning: the look-up lines were learned from all 39 days, these 8 stretches included. This check shows how well “stage + stack temperature” explains each reading. It is not a blind test like the stack temperature above.

The air heater (EAPH) box: computed vs shown

A closer look at one part, the 40 kW electric air heater that tops up the air before the stack (checked today, same 8 stretches, 97 h). **The twin computes the air leaving the heater well, but the numbers printed on the heater box come from the look-up, not from that computation.**

What	Stages	Twin vs plant	
Air into the stack, computed	Heat-up, hot standby	0.4–1.8 °C	Matches
Air into the stack, computed	Fuel switch, part load	4.7–8.0 °C	Matches
Air into the stack, computed	Air-flow cooling (heater off)	+39 °C	Differs
Heater outlet printed on the box	Heat-up air / N_2+H_2	27 / 18 °C	Close
Heater power printed on the box	Stage average	e.g. 20.4 vs 20.0 kW	Close
Heater power printed on the box	Ups and downs inside a stage	match 0.08 of 1	Differs

The printed power is one fixed value per stage (the plant median), while the real heater swings 14–26 kW while heating with N_2 + H_2. A test power from air flow × warming ÷ 0.89 follows those swings (match 0.94). With gas burning, though, it shows about 10 kW too much, because the twin's recuperator air is 35–73 °C too cold. Full check: air heater explainer.

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Scripts: session scratchpad cmp/ · replay.js (runs twin_2d/app.js: model, playbooks,
sensor readings), score.py, sens_extract.py, sens_score.py
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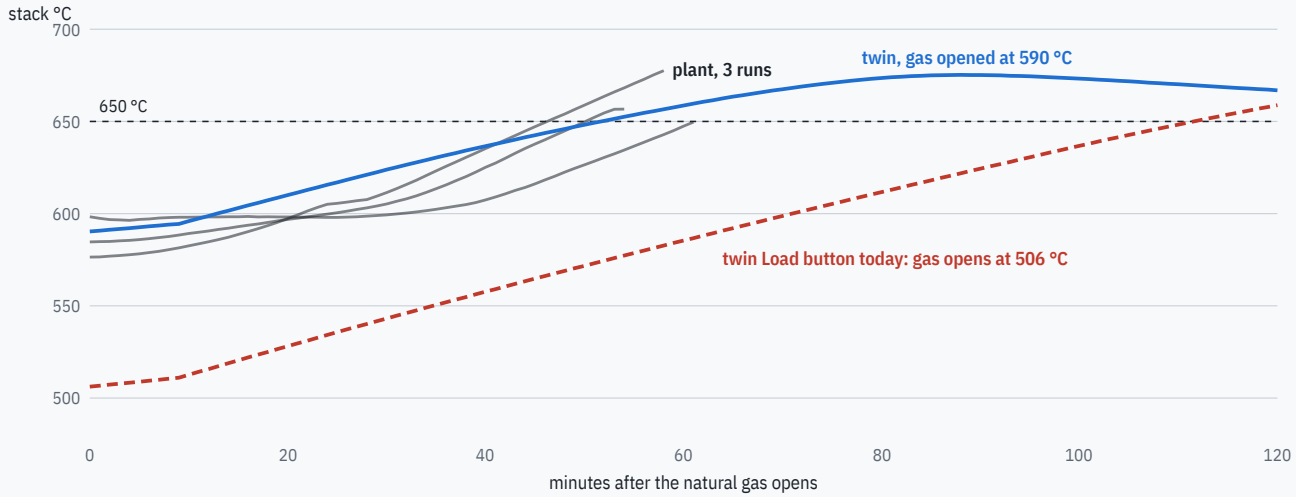
IF YOU REMEMBER ONE THING

Today's twin computes the stack, the incoming air and the stage name like the plant in every stage except fast forced cooling. Its other readings are typically within about 6 % of their range, but they are looked up, not computed, as the air heater box shows.

Today's twin: buttons and rules vs the plant

Most of the twin's automatic steps copy the plant, and since v15 the after-trip step does too: the twin goes back to hot standby, as the plant did 4 of 6 times. Two steps still differ. The Load button opens the gas from a 506 °C standby, while the plant first warms the stack to about 600 °C. And part load now always ends in hot standby, while the plant left part load in five different ways.

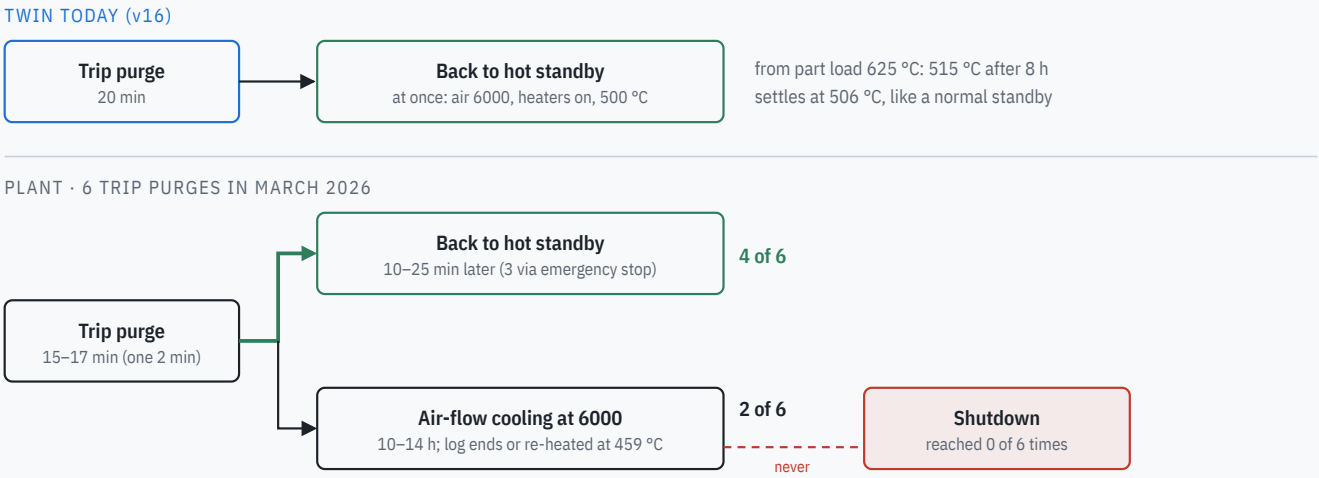
Load button: the gas opens about 90 °C too early



Grey: the 3 plant runs from hot standby to part load (gas opened at 577–598 °C; 650 °C after 47–51 min). Red dashed: the twin's Load button after a cold start, gas at 506 °C; it needs 112 min to reach 650 °C. Blue: the same button with the stack already at 590 °C; 650 °C after 52 min, like the plant. The heat balance is right. Only the starting point is wrong.

Across all 19 plant fuel switches the gas opened with the stack at 577–667 °C (median 600 °C). The plant had raised the air-heater target to 580–600 °C during hot standby, so the stack was already warm.

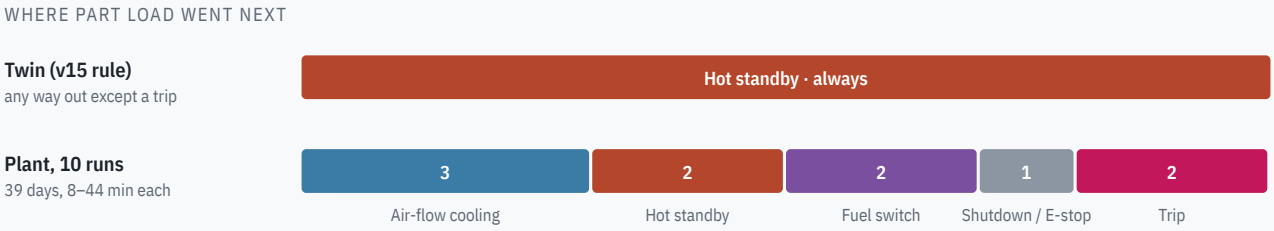
After a trip: twin and plant now take the same road



Twin v15 and later: after the 20-min purge the twin goes straight back to hot standby (this morning's v14 still cooled to shutdown, 39–44 h). In the plant data no trip ended in a full shutdown either. The only full cool-down after a trip came from an emergency stop without a purge: air 9000 ln/min, 664 → 41 °C in 18 h.

Twin test (v16): trip from part load at 625 °C → hot standby, 515 °C after 8 h, 506 °C after 3 days. Trip from hot standby at 511 °C → stays at 505–511 °C. The plant's way back was quicker and rougher: 3 of the 4 returns went through an emergency stop with the air off for 6–8 min.

Leaving part load: the twin is stricter than the plant



Since v15 the twin always unloads part load to hot standby, even when you press “Shut down / E-stop”. That shutdown then runs from hot standby. The plant went back to hot standby after only 2 of its 10 part-load runs; 3 went straight to air-flow cooling and 1 straight to shutdown.

Rule by rule

Button or rule	Twin today (v16)	Plant (39 days)	
Cold start	210 °C after 4.5 h, 300 °C 6.1 h, 490 °C 11.4 h; settles at 506 °C	4.7–5.0 h, 6.3–6.6 h, 11.6–12.0 h; settles at 509 °C	Matches
Load: when the gas opens	as soon as you press it (506 °C after a cold start)	stack at 577–667 °C, median 600 °C (19 fuel switches). The air-heater target is raised first.	Differs
Fuel switch → part load	fixed 10 min	median 9 min (2–38 min)	Matches
Part-load current	6.4 A per string (plant median)	2.7–13 A in the 3 replays	Close
Holding part load	keeper: above 660 °C more air, then cooler air-in; holds 655–661 °C	never longer than 44 min; hottest 677 °C	No plant data
Leaving part load (new, v15)	always to hot standby first: Shut down / E-stop, gas off or current 0 in part load all unload to standby (air 6000, heaters on, air heater 500 °C)	10 runs ended: 3 air-flow cooling, 2 hot standby, 2 back to fuel switch, 1 shutdown, 2 trips	Differs
Trip purge allowed	only from hot standby or part load; 20 min	5 of 6 purges came from those stages; 15–17 min	Matches
Automatic trip	stack safety sensor TIRZ 8.1.3 at 720 °C	that sensor read 609–696 °C at the 9 trips; other limits or the operator tripped the plant	No plant data
After the purge (changed, v15)	back to hot standby: air 6000, heaters on, air heater 500 °C. From part load 625 °C: 515 °C after 8 h, settles at 506 °C	4 of 6 back to hot standby in 10–25 min; 2 cooled 10–14 h; none reached shutdown	Matches
Emergency stop (renamed, v16)	shown as the air-off stage “Shutdown / Emergency stop”; the “Shut down / E-stop” button cools at 8000 ln/min, air off below 100 °C	3 emergency stops without purge (in part load, air-flow cooling, fuel switch); 3 more after purges cut the air for 6–8 min, then back to standby	Differs

Button or rule	Twin today (v16)	Plant (39 days)	
Cooling at 6000 ln/min	8 h: 472 °C · 14 h: 345 °C (same physics; after a trip the twin no longer does this)	after trips #1 and #3: 8 h: 485 °C · 14 h: 461 °C	Close
Air off at the end of a cool-down	below 100 °C	25–46 °C on the full cool-downs	Differs

Cooling at 6000: twin started from the plant's own temperature at each purge's end. Plant trip details: 9 trips in March 2026 (6 purges, 3 emergency stops without purge).

Decided today

After a trip → back to hot standby (v15). Replaces “always cool to shutdown”. Now matches the plant in 4 of 6 purges.

Part load → always hot standby (v15) and the name **Shutdown / Emergency stop** (v16). Both are your rules; the first is stricter than the plant (2 of 10 runs).

Decisions for you

A. Load button. Make it first raise the air-heater target to 580–600 °C and open the gas only when the stack reaches about 590 °C, as the plant does? Or keep “gas at once”? Suggested: follow the plant.

B. Leaving part load (new). Keep “always hot standby first”, or also allow part load straight to air-flow cooling or shutdown, as the plant did 4 of 10 times?

C. Fast cooling at 9000 air. Let the heat recovery drop to about 78 % at 9000 ln/min (passes, but tuned on one stretch), or wait for more forced-cooling days?

D. Small timings. Purge 20 → 15 min, and air off at about 40 °C instead of 100 °C, to match the plant?

E. Pass mark. Six of seven stages are now within 5 °C. Tighten the stack pass mark from ±20 to ±10 °C?

F. Air heater box (new). Print the twin's own computed air temperature on the EAPH box, and later its power from air flow × warming ÷ 0.89 once the recuperator is fixed? (Decisions A–D in the air heater explainer.)

G. Emergency stop (new). The twin's “Shut down / E-stop” is a slow cool-down. Should a real emergency stop cut the air at once, as the plant did (6–8 min air off)?

IF YOU REMEMBER ONE THING

The twin's physics matches the plant, and after a trip it now returns to hot standby like the plant. What still differs: the plant warms the standby to about 600 °C before it opens the gas, and it leaves part load in more ways than the twin allows.